



Antenna Datasheet

Product OC: YF0023AA

Version: 2.0

Date: 2023-06-13

Status: Released

Product Name: Wi-Fi FPC Antenna

Key Features:

Frequency band: 2400–2500 MHz, 5150–5850 MHz, 5925–7125 MHz

Peak efficiency: 80.08 % (FS)

Dimensions: 22.9 × 11.7 mm

RoHS and REACH Compliant

Overview

Quectel Wi-Fi antenna covers 2.4 GHz, 5 GHz, and up to 7 GHz bands, fully satisfying customers' requirements for Wi-Fi 5, Wi-Fi 6, and Wi-Fi 6E/Wi-Fi 7. There are various antenna types, including built-in FPC/PCB antenna, ceramic patch antenna, and other external antennas of different shapes or sizes. The antenna performance meets the customers' demands for efficiency, gain, and radiation and ensures the superior experience of the customers' products in use.

Contents

Overview.....	1
Contents	2
1 Specification.....	3
1.1. Electrical.....	3
1.2. Mechanical, Environmental & Storage.....	4
2 Drawing	5
3 Detailed Performance	6
3.1. S-Parameter Test	6
3.1.1. VSWR.....	6
3.1.2. Return Loss	7
3.2. Radiation Performance Test.....	8
3.2.1. Efficiency	8
3.2.2. Average Gain	9
3.2.3. Peak Gain.....	10
3.2.4. 3D & 2D Radiation Pattern.....	11
3.2.4.1. Test Condition: Free Space	11
3.2.4.2. Test Condition: Stick on 2 mm Thick ABS Board	13
4 Packaging	15
Contact Us.....	17
Legal Notices	18
Revision History	20

1 Specification

Test Condition: Free Space & Stick on 2 mm Thick ABS Board

1.1. Electrical

Electrical	
Frequency Range	2400–2500 MHz, 5150–5850 MHz, 5925–7125 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

Specification	Band	Band	Wi-Fi 2G	Wi-Fi 5G	Wi-Fi 7G
		Freq. (MHz)	2400 - 2500	5150 - 5850	5925 - 7125
Max. VSWR	FS		6.1	2.8	1.4
	ABS		3.3	1.4	1.5
Max. Return Loss (dB)	FS		-2.9	-6.6	-15.0
	ABS		-5.4	-16.1	-13.5
AVG Eff. (%)	FS		30.1	54.7	67.5
	ABS		53.6	65.5	59.6
AVG Gain (dB)	FS		-5.2	-2.7	-1.7
	ABS		-2.7	-1.8	-2.3
Max. Peak Gain (dBi)	FS		0.2	2.9	4.2
	ABS		2.7	4.3	6.3
VSWR	FS		≤ 6.1		
	ABS		≤ 3.3		

Return Loss	FS	≤ -2.9 dB
	ABS	≤ -5.4 dB
Peak Gain	FS	≤ 4.2 dBi
	ABS	≤ 6.3 dBi

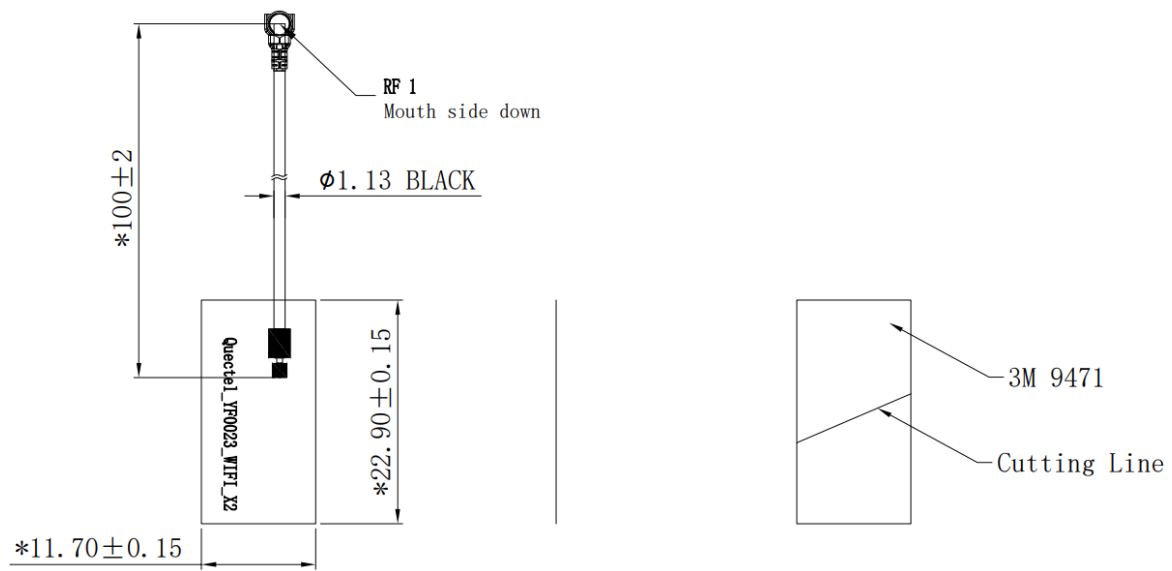
- FS: Free Space
- ABS: Stick on 2 mm Thick ABS Board

1.2. Mechanical, Environmental & Storage

Mechanical	
Antenna Size	22.9 × 11.7 mm
Antenna Material & Color	FPC & Black
Cable Type & Color & Length	Φ 1.13 & Black & 100 mm
Connector Type	RF I
Mounting Type	Adhesive
Antenna Weight	Typ. 0.4 g
Environmental	
Operation Temperature	-20 °C to +85 °C
RoHS and REACH Compliant	Yes

Storage	
Storage Temperature	18 °C–27 °C
Humidity	30 %–80 % RH
Storage Place	Away from corrosive gas and direct sunlight.
Packaging	Antennas should be stored in unopened sealed manufacturer's plastic packaging.

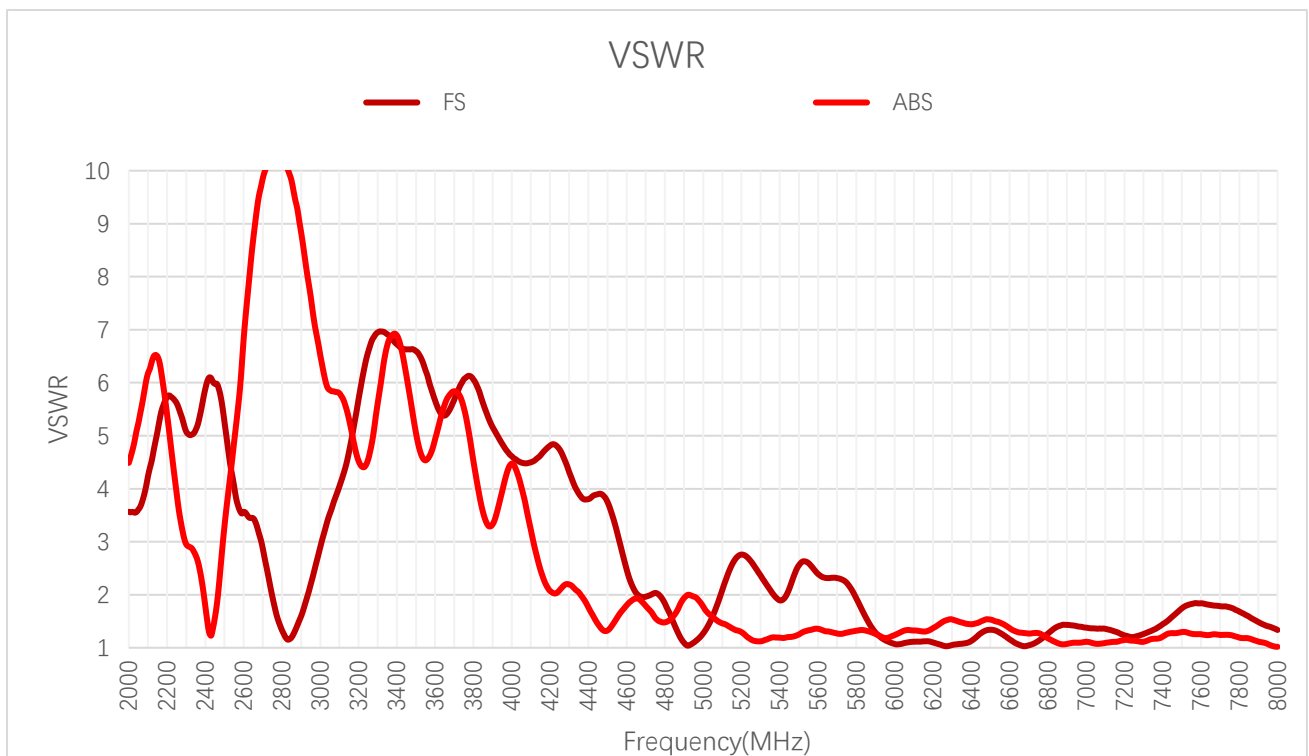
2 Drawing



3 Detailed Performance

3.1. S-Parameter Test

3.1.1. VSWR



VSWR

Frequency (MHz)	2400	2450	2500	5150	5500	5850	5920	6520	7120
FS	5.9	6.0	5.2	2.6	2.6	1.6	1.2	1.3	1.3
ABS	1.8	1.6	3.3	1.4	1.3	1.3	1.2	1.5	1.1

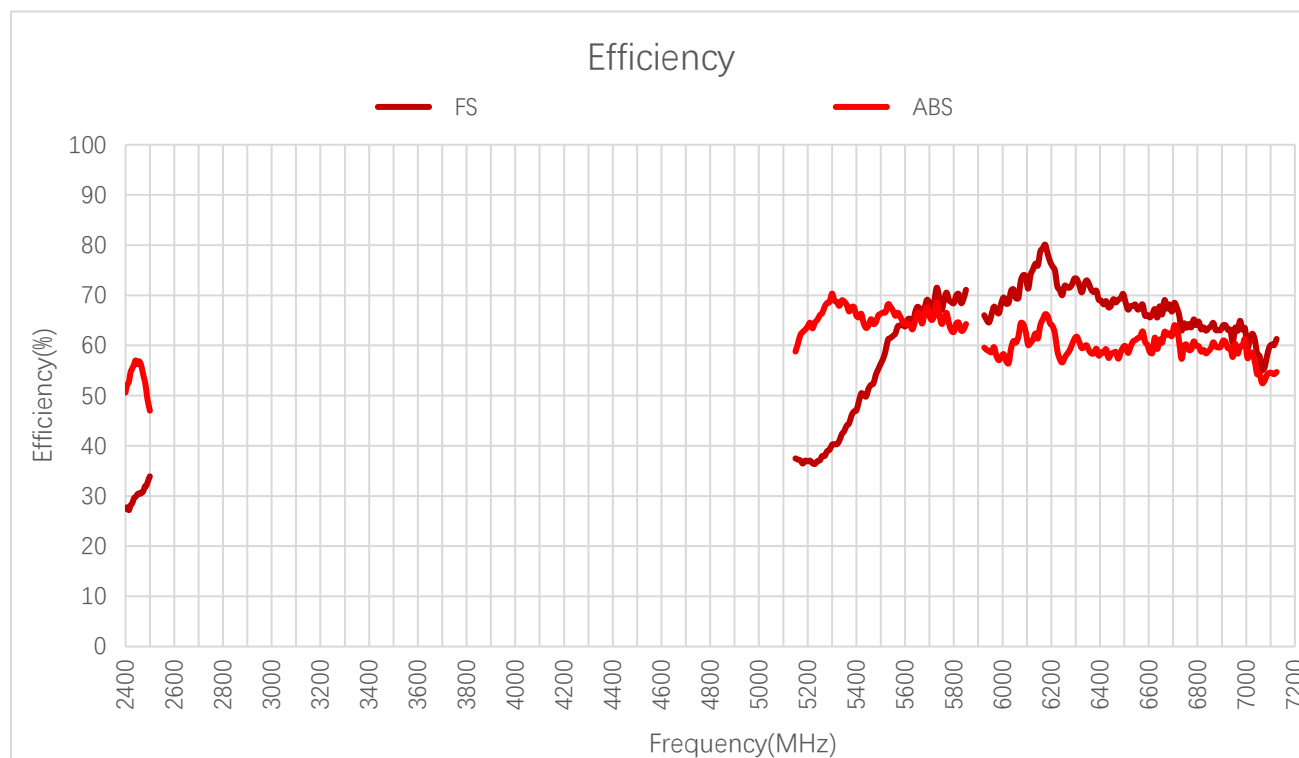
3.1.2. Return Loss



Return Loss (dB)									
Frequency (MHz)	2400	2450	2500	5150	5500	5850	5920	6520	7120
FS	-3.0	-2.9	-3.4	-7.2	-7.2	-13.1	-20.1	-17.0	-16.7
ABS	-11.1	-13.0	-5.4	-16.1	-19.1	-17.2	-19.8	-13.9	-26.3

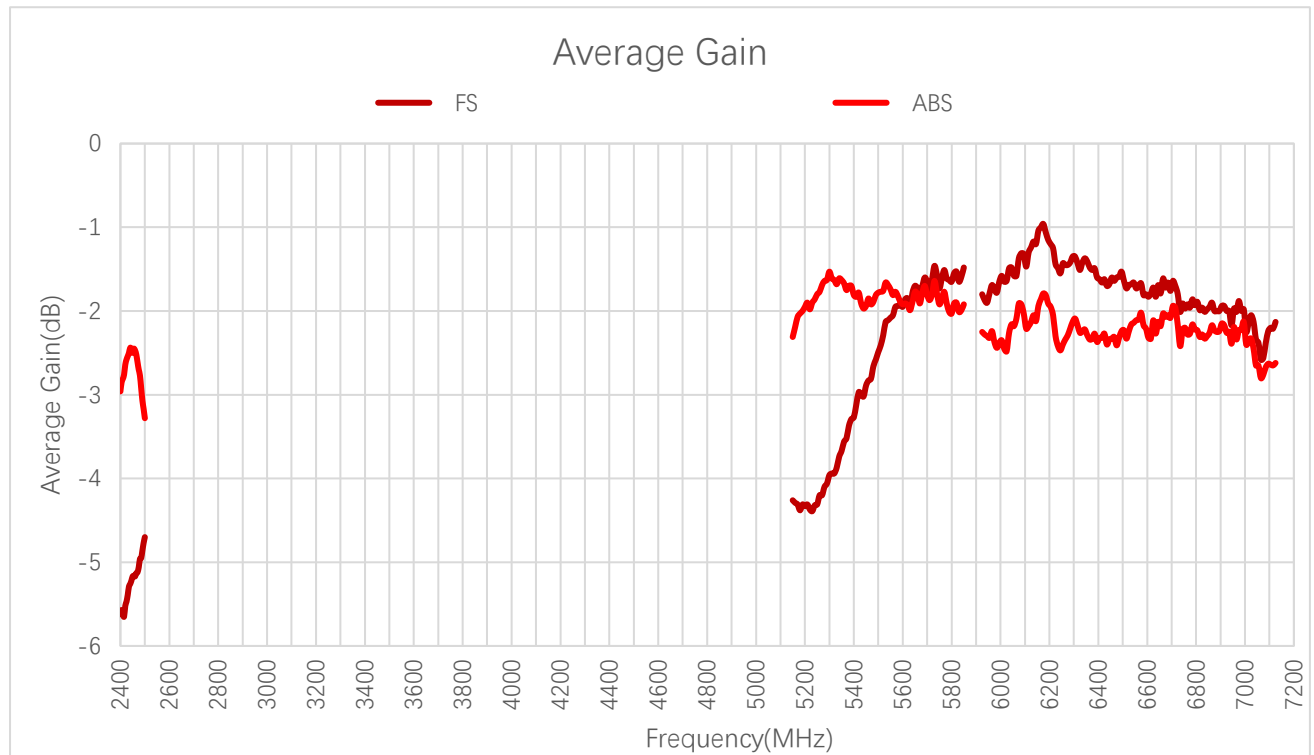
3.2. Radiation Performance Test

3.2.1. Efficiency



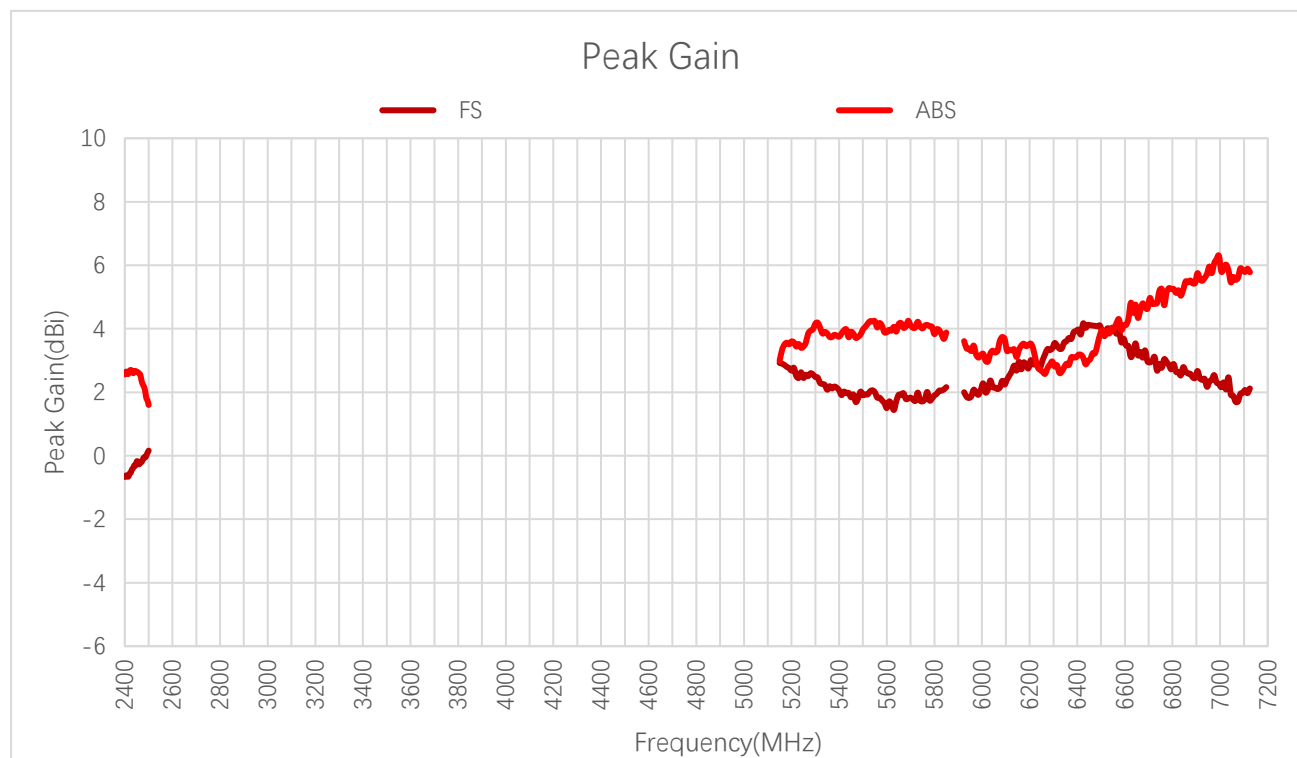
Efficiency (%)									
Frequency (MHz)	2400	2450	2500	5150	5500	5850	5925	6525	7125
FS	27.5	30.4	33.9	37.5	56.4	71.1	66.0	67.8	61.3
ABS	50.6	56.3	47.0	58.8	66.3	64.3	59.6	59.7	54.7

3.2.2. Average Gain



Average Gain (dB)									
Frequency (MHz)	2400	2450	2500	5150	5500	5850	5925	6525	7125
FS	-5.6	-5.2	-4.7	-4.3	-2.5	-1.5	-1.8	-1.7	-2.1
ABS	-3.0	-2.5	-3.3	-2.3	-1.8	-1.9	-2.3	-2.2	-2.6

3.2.3. Peak Gain



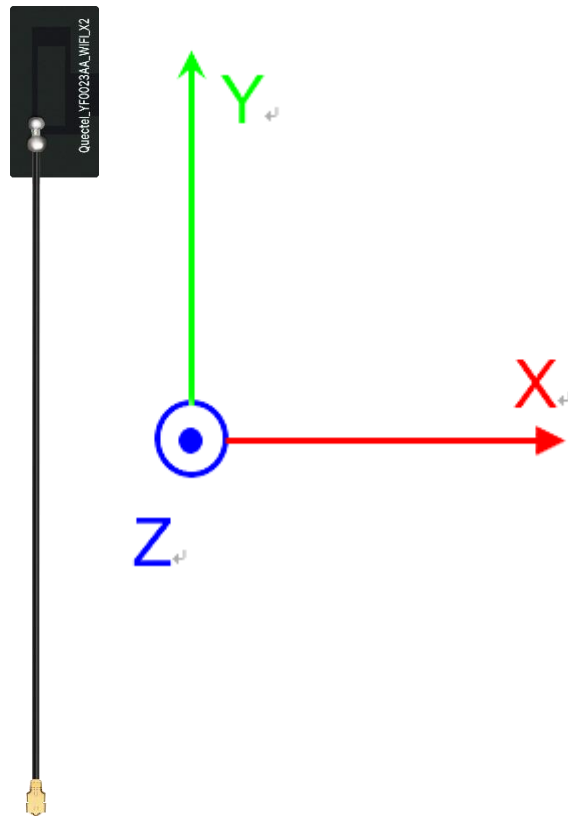
Peak Gain (dBi)

Frequency (MHz)	2400	2450	2500	5150	5500	5850	5925	6525	7125
FS	-0.7	-0.2	0.2	2.9	1.9	2.2	2.0	4.0	2.1
ABS	2.6	2.6	1.6	3.0	4.0	3.9	3.6	3.9	5.8

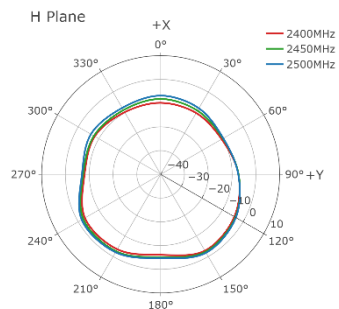
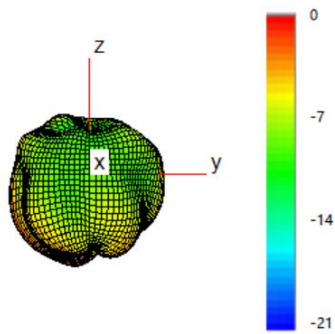
3.2.4. 3D & 2D Radiation Pattern

3.2.4.1. Test Condition: Free Space

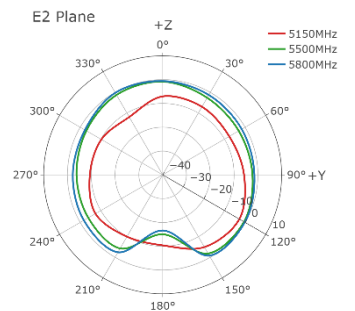
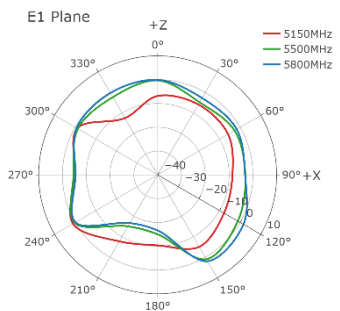
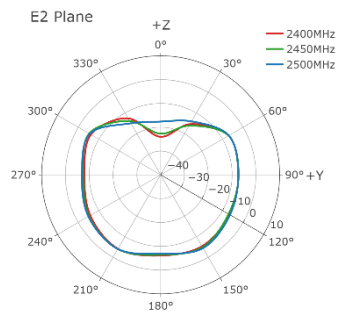
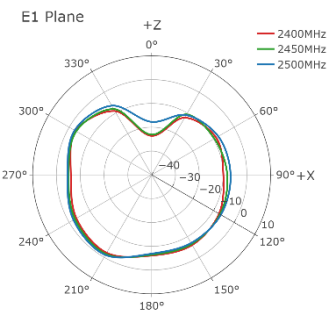
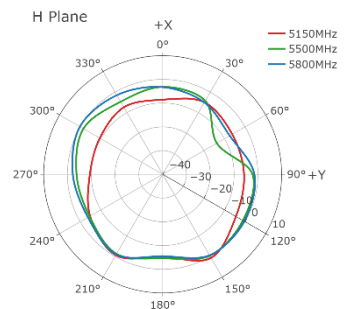
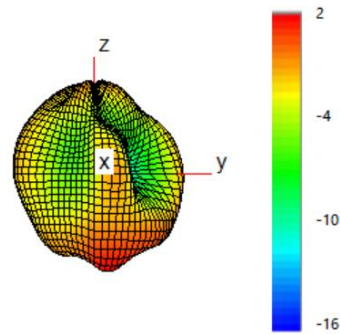
- Test Chamber: HF-G-1



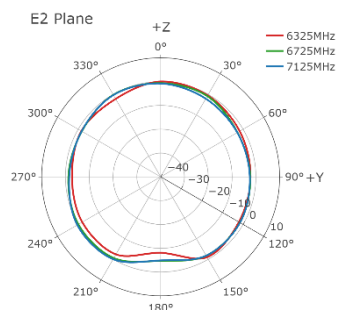
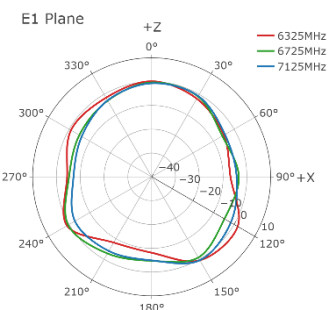
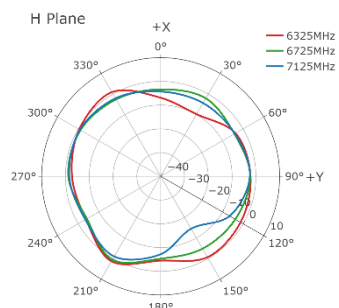
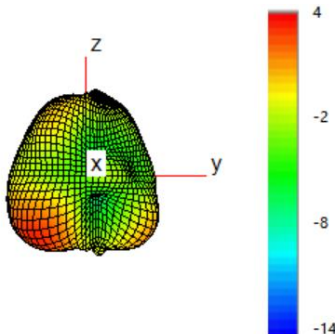
2450 MHz



5500 MHz

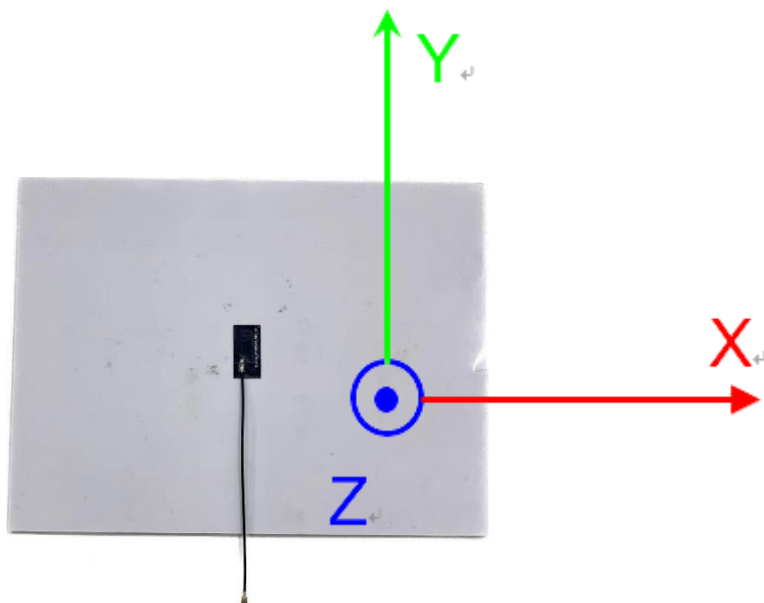


6725 MHz

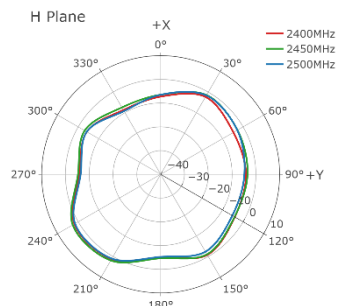
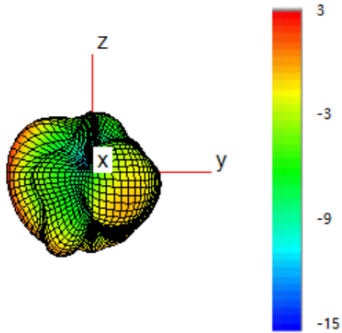


3.2.4.2. Test Condition: Stick on 2 mm Thick ABS Board

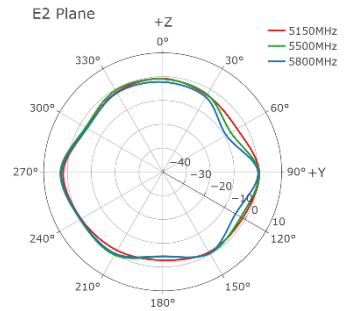
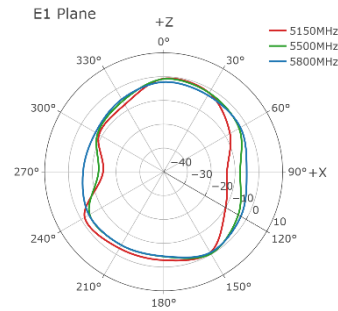
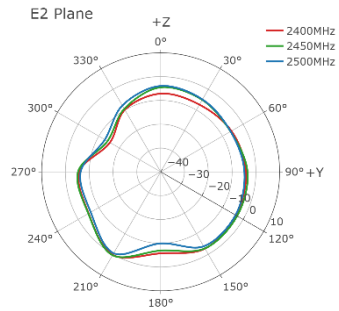
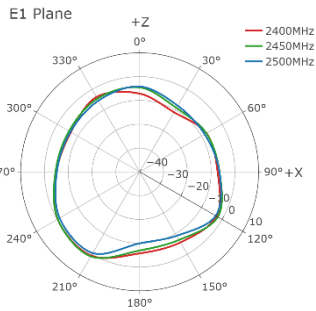
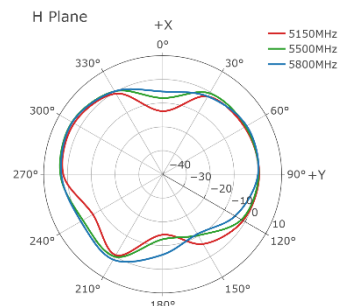
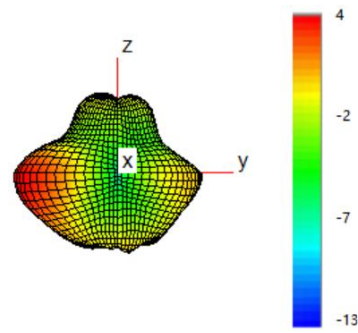
- Test Chamber: HF-G-1



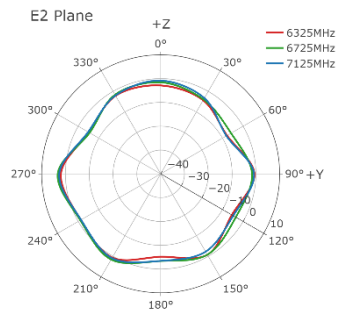
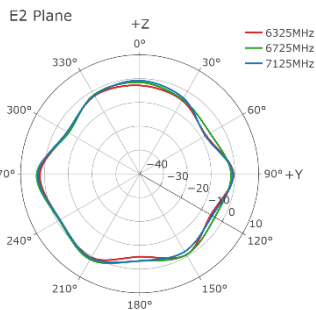
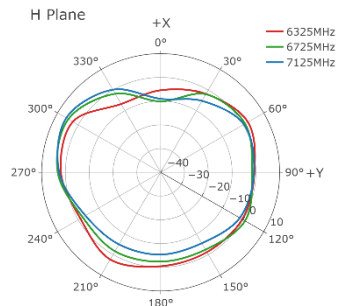
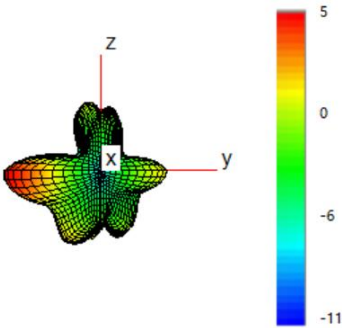
2450 MHz



5500 MHz

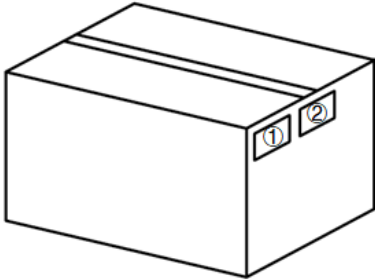
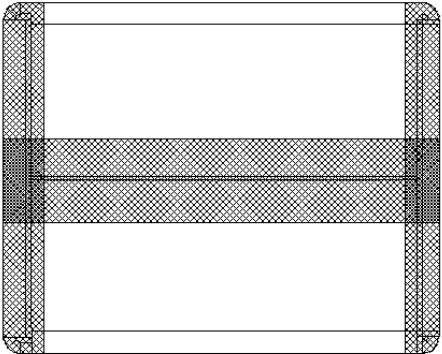


6725 MHz



4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		50 products in a bundle
2		200 pcs antenna products in a PE bag; (200 pcs antennas per PE bag)
3		The quantity of the full container depends on the actual packing; <u>Carton Size:</u> <u>L × W × H = 390 × 270 × 150 mm</u>

4		Position for Attaching Labels ① Carton Label ② Quality Label
5		Sealing Cartons “Γ” type sealing cartons

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties (“third-party materials”). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2023. All rights reserved.

Revision History

Version	Date	Author	Note
-	2021-08-10	Toby WANG	Creation of the document
1.0	2021-08-10	Toby WANG	First official release
1.1	2021-11-30	Toby WANG	Updated the product description (Chapter 1).
2.0	2023-06-13	Lucky FENG/ David LIU/ Vinnie LIU	Updated the template and all test data.



www.quectel.com